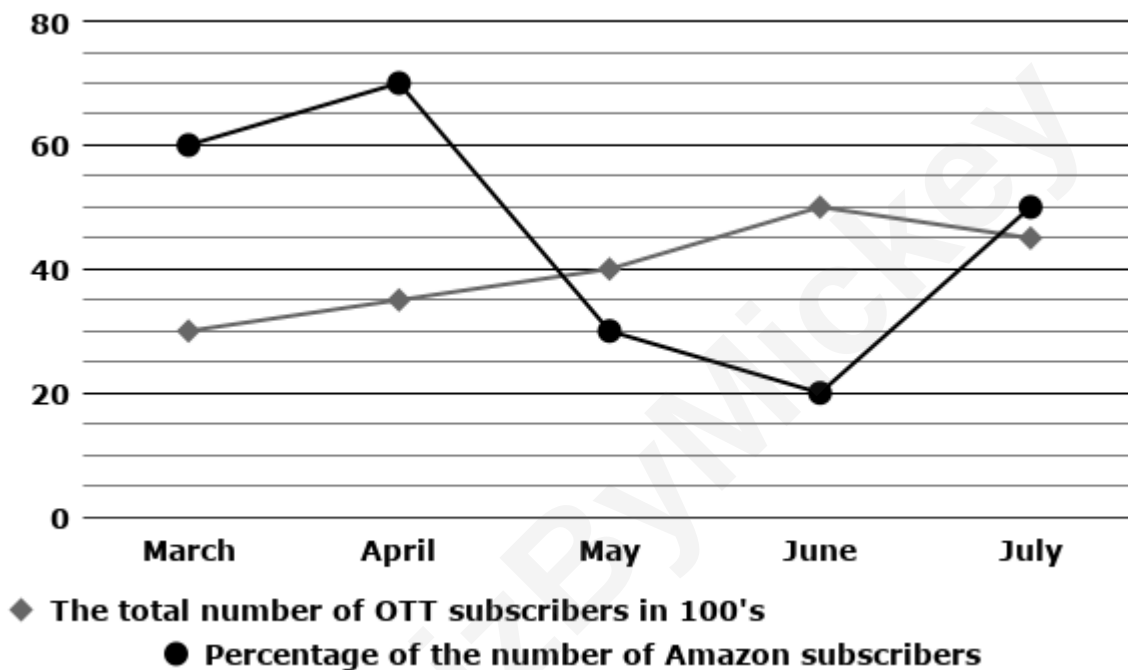


1. Questions

Study the following information carefully and answer the following questions.

The given line graph shows the total number of OTT subscribers in the 100's in five different months, namely March, April, May, June, and July, respectively, and also the percentage of the number of Amazon subscribers in these five different months.

Note: The total number of OTT subscribers in each month = The number of Amazon subscribers + The number of Netflix subscribers.



The total number of OTT subscribers in January is $\frac{3}{7}$ th more than that in April, and the ratio of the number of Amazon subscribers in March to January is 6:7. Find the number of Netflix subscribers in January.

- 3200
- 2450
- 2100
- 2900
- 3000

2. Questions

In June, the ratio of the number of males in Amazon to Netflix subscribers is 1:5 and the ratio of the number of females in Amazon to Netflix subscribers is 1:3. Find the number of male Amazon subscribers.

- 280
- 400

- c. 700
- d. 500
- e. 800

3. Questions

In May, the number of Zee5 subscribers was 25% more than that of Amazon subscribers. Find the difference between the number of Netflix and Zee5 subscribers.

- a. 1300
- b. 1100
- c. 900
- d. 800
- e. 1500

4. Questions

Find the difference between the sum of the number of Amazon subscribers in March and April together and the number of Netflix subscribers in July.

- a. 1800
- b. 2000
- c. 2100
- d. 1560
- e. 2400

5. Questions

Find the ratio of the total number of subscribers in June to the number of Amazon subscribers in July.

- a. 20: 9
- b. 11: 12
- c. 18: 19
- d. 21: 23
- e. 14: 15

6. Questions

Study the following information carefully and answer the following questions.

The given table chart shows the total number of students in five different colleges namely A, B, C, D and E respectively and the difference between the number of boys and girls in these different colleges.

Note: The number of boys in each college is more than that of girls.

Colleges	The total number of students	The difference between the number of boys and girls
A	480	80
B	560	20
C	380	20
D	440	40
E	500	60

The total number of students in college A is 20% less than that in college F and the average number of boys in college B and F is 250. Find the number of girls in college F.

- a. 390
- b. 420
- c. 450
- d. 290
- e. 330

7. Questions

If 20% of the boys in college C are transferred to college D and $\frac{1}{6}$ th of the girls in college C are transferred to college D, then find the total number of students in college D after transferring.

- a. 480
- b. 520
- c. 510
- d. 600
- e. 490

8. Questions

In college A, out of the number of boys, 40% stay in hostels. If the number of girls who stay in a hostel is 50 less than that of boys, then find the sum of the number of boys and girls who stay in the hostel.

- a. 188

- b. 174
- c. 201
- d. 145
- e. 215

9. Questions

Find the sum of the number of girls in colleges A, C, and D together.

- a. 580
- b. 420
- c. 320
- d. 610
- e. 600

10. Questions

The total number of students in college E is what percentage of the difference between the number of girls in colleges C and D?

- a. 1800%
- b. 2200%
- c. 1200%
- d. 2500%
- e. 1400%

11. Questions

If the sum of the number of boys in colleges A and B is $(2x + 80)$, then find the value of $8x$.

- a. 2180
- b. 1960
- c. 2200
- d. 1450
- e. 1000

12. Questions

45 men complete 60% of the work in 12 days working 5 hours in a day. If after 10 days, only 20% of the work is completed, then how many additional hours are required to complete the work on time and without increasing the number of men.

- a. 5
- b. 3
- c. 7
- d. 8
- e. 1

13. Questions

In a family, a couple has a daughter and son. The age of the father is 'm' times that of his son and the age of the daughter is half of his mother's. The wife is 20 years younger than her husband, and the sister is seven years older than her brother. Find what will be the age of the mother.

- a. 45 years
- b. 40 years
- c. 35 years
- d. 50 years
- e. Cannot be determined

14. Questions

688 liters of mixture contains juice and water in the ratio 3:1. If 64 liters of mixture is replaced with 32 liters of juice and 4 liters of water, then the ratio of the quantity of juice to water becomes 25:x. Find the value of x.

- a. 4
- b. 8
- c. 2
- d. 11
- e. 13

15. Questions

A and B invested Rs. $(x + 1500)$ and Rs. $(2x + 600)$, respectively. After 18 months, A withdrew 20% of his investment, while B withdrew Rs. $(x + 450)$ from his investment. If the ratio of the profits received by A and B after 2 years is 152:109, then find the value of x.

- a. 800
- b. 500
- c. 400
- d. 1200
- e. 1100

16. Questions

Two boats, A and B, are at opposite ends of a river. The speeds of boats A and B are 30 kmph and 20 kmph respectively. The speed of the stream is 10 kmph, which flows from the boat B to boat A side. If both boats start moving towards each other and meet after 3 hours, then find the total length of the river.

- a. 200 km
- b. 150 km
- c. 250 km
- d. 300 km
- e. 450 km

17. Questions

Mahi had Rs. 40000 with him. He invested 30% of the amount in scheme A at the rate of 20% per annum and the rest of the amount in scheme B at the rate of 15% per annum. Find the simple interest received in both schemes after 2 years.

- a. Rs. 13200
- b. Rs. 21800
- c. Rs. 8800
- d. Rs. 11500
- e. Rs. 7200

18. Questions

The volume of the cylinder is 2310 cm^3 and the height of the cylinder is 15 cm. If the side of the square is 3 cm more than the radius of the cylinder, then find the perimeter of the square.

- a. 80 cm
- b. 120 cm
- c. 40 cm
- d. 64 cm
- e. 88 cm

19. Questions

The ratio between the two numbers is 5:3 respectively. The difference between the squares of the two numbers is equal to 1600. If the larger number is increased by 300%, then find the resultant value.

- a. 350

- b. 280
- c. 400
- d. 200
- e. 500

20. Questions

A shopkeeper marked an article by 50% above its cost price and sold it after two successive discounts of 15% and 20%, respectively. If the shopkeeper had sold the article after giving a single discount of 10%, then he would earn Rs. 99 more. Find the cost price of the article.

- a. Rs. 300
- b. Rs. 450
- c. Rs. 400
- d. Rs. 200
- e. Rs. 500

21. Questions

What value should come in the place of (?) in the following questions?

$$18\% \text{ of } 800 + 42 * 3 - 8^2 = ? * 2$$

- a. 125
- b. 215
- c. 103
- d. 100
- e. 98

22. Questions

$$18^2 - 3^3 + 5^2 * 3 + ? = 90 * 4$$

- a. -22
- b. -10
- c. -8
- d. -15
- e. -12

23. Questions

$$178 + 30 * 8/15 - 20 * 3/4 = ?/2$$

- a. 300
- b. 358
- c. 288
- d. 201
- e. 562

24. Questions

$$(256)^{1/2} * 4^{1/2} + 31 * 8 = ?^2 - 9^2$$

- a. 22
- b. 17
- c. 13
- d. 19
- e. 10

25. Questions

$$12 * 17 + 240 * 5/8 = ? + 190$$

- a. 172
- b. 164
- c. 201
- d. 255
- e. 188

26. Questions

What approximate value should come in the place of (?) in the following questions?

$$299.89 * 29.89 - (24.12\% \text{ of } 1199.79) = ? * 2.12$$

- a. 4356
- b. 2810
- c. 3890
- d. 4200
- e. 1560

27. Questions

$$15^2 + 15.12 * 2.87 - ? = 99.12\% \text{ of } 200.21$$

- a. 90
- b. 78
- c. 72
- d. 102
- e. 67

28. Questions

$$1400.01 + \sqrt{1600} - ? = 33.12\% \text{ of } 700.12$$

- a. 1880
- b. 1109
- c. 1209
- d. 890
- e. 980

29. Questions

$$167.12 - 78.09 + 5.12^2 * 4.98 = ?^2 - 11.12$$

- a. 18
- b. 22
- c. 21
- d. 15
- e. 20

30. Questions

$$42.12 * 19.12/7 = 8^{1/3} + 156.98 - ?$$

- a. 55
- b. 40
- c. 23
- d. 90
- e. 45

31. Questions

What value should come in the place of (?) in the following number series?

?, 16, 50, 202, 1012, 6074

- a. 3
- b. 2
- c. 10
- d. 12
- e. 7

32. Questions**121, 152, ?, 230, 273, 320**

- a. 189
- b. 165
- c. 187
- d. 157
- e. 140

33. Questions**24, 23, ?, 129, 512**

- a. 34
- b. 44
- c. 40
- d. 38
- e. 89

34. Questions**19, 10, 11, ?, 38, 97.5**

- a. 18
- b. 16
- c. 22
- d. 26
- e. 25

35. Questions**146, 171, 198, 227, 258, ?**

- a. 265

- b. 291
- c. 332
- d. 415
- e. 301

36. Questions

Find out the wrong number in the following number series.

188, 204, 224, 260, 308, 368

- a. 368
- b. 224
- c. 188
- d. 308
- e. 204

37. Questions

79, 73, 700, 689, 8268, 8255

- a. 700
- b. 689
- c. 8268
- d. 8255
- e. 73

38. Questions

648, 216, 76, 24, 8

- a. 24
- b. 76
- c. 8
- d. 648
- e. 216

39. Questions

517, 738, 1005, 1336, 1733

- a. 1005

- b. 1336
- c. 1733
- d. 517
- e. 738

40. Questions**1245, 1249, 1259, 1273, 1305**

- a. 1273
- b. 1305
- c. 1245
- d. 1259
- e. 1249

41. Questions

The following question contains two equations, as I and II. You have to solve both equations and determine the relationship between them and give the answer as

i). $x^2 - 27x + 182 = 0$

ii). $y^2 - 24y + 143 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or the relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

42. Questions

i). $3x^2 + 32x + 84 = 0$

ii). $y^2 + 15y + 56 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or the relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

43. Questions

i). $x^2 - 24x + 135 = 0$

ii). $y^2 - 33y + 272 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or the relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

44. Questions

i). $x^2 + 17x + 60 = 0$

ii). $y^2 + 9y + 20 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or the relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

45. Questions

i). $x^2 - 12x + 36 = 0$

ii). $3y^2 - 19y + 20 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or the relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

Explanations:**1. Questions****In March:**

The total number of OTT subscribers = 3000

The number of Amazon subscribers = $3000 * 60/100 = 1800$

The number of Netflix subscribers = $3000 * 40/100 = 1200$

In April:

The total number of OTT subscribers = 3500

The number of Amazon subscribers = $3500 * 70/100 = 2450$

The number of Netflix subscribers = $3500 * 30/100 = 1050$

In May:

The total number of OTT subscribers = 4000

The number of Amazon subscribers = $4000 * 30/100 = 1200$

The number of Netflix subscribers = $4000 * 70/100 = 2800$

In June:

The total number of OTT subscribers = 5000

The number of Amazon subscribers = $5000 * 20/100 = 1000$

The number of Netflix subscribers = $5000 * 80/100 = 4000$

In July:

The total number of OTT subscribers = 4500

The number of Amazon subscribers = $4500 * 50/100 = 2250$

The number of Netflix subscribers = $4500 * 50/100 = 2250$

Months	The total number of OTT subscribers	The number of Amazon subscribers	The number of Netflix subscribers
March	3000	1800	1200
April	3500	2450	1050
May	4000	1200	2800
June	5000	1000	4000
July	4500	2250	2250

Answer: D

The total number of OTT subscribers in January = $(3500 + 3500 * 3/7) = (3500 + 1500) = 5000$

The number of Amazon subscribers in January = $1800 * 7/6 = 2100$

The number of Netflix subscribers in January = $(5000 - 2100) = 2900$

Hence, the correct answer is option D.

2. Questions

In March:

The total number of OTT subscribers = 3000

The number of Amazon subscribers = $3000 * 60/100 = 1800$

The number of Netflix subscribers = $3000 * 40/100 = 1200$

In April:

The total number of OTT subscribers = 3500

The number of Amazon subscribers = $3500 * 70/100 = 2450$

The number of Netflix subscribers = $3500 * 30/100 = 1050$

In May:

The total number of OTT subscribers = 4000

The number of Amazon subscribers = $4000 * 30/100 = 1200$

The number of Netflix subscribers = $4000 * 70/100 = 2800$

In June:

The total number of OTT subscribers = 5000

The number of Amazon subscribers = $5000 * 20/100 = 1000$

The number of Netflix subscribers = $5000 * 80/100 = 4000$

In July:

The total number of OTT subscribers = 4500

The number of Amazon subscribers = $4500 * 50/100 = 2250$

The number of Netflix subscribers = $4500 * 50/100 = 2250$

Months	The total number of OTT subscribers	The number of Amazon subscribers	The number of Netflix subscribers
March	3000	1800	1200
April	3500	2450	1050
May	4000	1200	2800
June	5000	1000	4000
July	4500	2250	2250

Answer: D

Let, the number of males in Amazon to Netflix subscribers in June is x and $5x$ respectively.

Let, the number of females in Amazon to Netflix subscribers in June is y and $3y$ respectively.

$$x + y = 1000 \text{ ----> (1)}$$

$$5x + 3y = 4000 \text{ ----> (2)}$$

Multiplying (1) * 5 and (2) * 1, we get

$$5x + 5y = 5000 \text{ ----> (3)}$$

$$5x + 3y = 4000 \text{ -----> (4)}$$

By solving equation (3) and (4), we get

$$x = 500 \text{ and } y = 500$$

The number of male Amazon subscribers in June = 500

Hence, the correct answer is option D.

3. Questions

In March:

The total number of OTT subscribers = 3000

The number of Amazon subscribers = $3000 * 60/100 = 1800$

The number of Netflix subscribers = $3000 * 40/100 = 1200$

In April:

The total number of OTT subscribers = 3500

The number of Amazon subscribers = $3500 * 70/100 = 2450$

The number of Netflix subscribers = $3500 * 30/100 = 1050$

In May:

The total number of OTT subscribers = 4000

The number of Amazon subscribers = $4000 * 30/100 = 1200$

The number of Netflix subscribers = $4000 * 70/100 = 2800$

In June:

The total number of OTT subscribers = 5000

The number of Amazon subscribers = $5000 * 20/100 = 1000$

The number of Netflix subscribers = $5000 * 80/100 = 4000$

In July:

The total number of OTT subscribers = 4500

The number of Amazon subscribers = $4500 * 50/100 = 2250$

The number of Netflix subscribers = $4500 * 50/100 = 2250$

Months	The total number of OTT subscribers	The number of Amazon subscribers	The number of Netflix subscribers
March	3000	1800	1200
April	3500	2450	1050
May	4000	1200	2800
June	5000	1000	4000
July	4500	2250	2250

Answer: A

The number of Zee5 subscribers in May = $1200 * 125/100 = 1500$

The number of Netflix subscribers in May = 2800

Required difference = $(2800 - 1500) = 1300$

Hence, the correct answer is option A.

4. Questions**In March:**

The total number of OTT subscribers = 3000

The number of Amazon subscribers = $3000 * 60/100 = 1800$

The number of Netflix subscribers = $3000 * 40/100 = 1200$

In April:

The total number of OTT subscribers = 3500

The number of Amazon subscribers = $3500 * 70/100 = 2450$

The number of Netflix subscribers = $3500 * 30/100 = 1050$

In May:

The total number of OTT subscribers = 4000

The number of Amazon subscribers = $4000 * 30/100 = 1200$

The number of Netflix subscribers = $4000 * 70/100 = 2800$

In June:

The total number of OTT subscribers = 5000

The number of Amazon subscribers = $5000 * 20/100 = 1000$

The number of Netflix subscribers = $5000 * 80/100 = 4000$

In July:

The total number of OTT subscribers = 4500

The number of Amazon subscribers = $4500 * 50/100 = 2250$

The number of Netflix subscribers = $4500 * 50/100 = 2250$

Months	The total number of OTT subscribers	The number of Amazon subscribers	The number of Netflix subscribers
March	3000	1800	1200
April	3500	2450	1050
May	4000	1200	2800
June	5000	1000	4000
July	4500	2250	2250

Answer: B

The sum of the number of Amazon subscribers in March and April together = $(1800 + 2450) = 4250$

The number of Netflix subscribers in July = 2250

Required difference = $(4250 - 2250) = 2000$

Hence, the correct answer is option B.

5. Questions

In March:

The total number of OTT subscribers = 3000

The number of Amazon subscribers = $3000 * 60/100 = 1800$

The number of Netflix subscribers = $3000 * 40/100 = 1200$

In April:

The total number of OTT subscribers = 3500

The number of Amazon subscribers = $3500 * 70/100 = 2450$

The number of Netflix subscribers = $3500 * 30/100 = 1050$

In May:

The total number of OTT subscribers = 4000

The number of Amazon subscribers = $4000 * 30/100 = 1200$

The number of Netflix subscribers = $4000 * 70/100 = 2800$

In June:

The total number of OTT subscribers = 5000

The number of Amazon subscribers = $5000 * 20/100 = 1000$

The number of Netflix subscribers = $5000 * 80/100 = 4000$

In July:

The total number of OTT subscribers = 4500

The number of Amazon subscribers = $4500 * 50/100 = 2250$

The number of Netflix subscribers = $4500 * 50/100 = 2250$

Months	The total number of OTT subscribers	The number of Amazon subscribers	The number of Netflix subscribers
March	3000	1800	1200
April	3500	2450	1050
May	4000	1200	2800
June	5000	1000	4000
July	4500	2250	2250

Answer: A

The total number of subscribers in June = 5000

The number of Amazon subscribers in July = 2250

Required ratio = 5000: 2250 = 100: 45 = 20: 9

Hence, the correct answer is option A.

6. Questions

In college A:

The total number of students = 480

The number of boys = $(480 + 80)/2 = 280$

The number of girls = $(480 - 280) = 200$

In college B:

The total number of students = 560

The number of boys = $(560 + 20)/2 = 290$

The number of girls = $(560 - 290) = 270$

In college C:

The total number of students = 380

The number of boys = $(380 + 20)/2 = 200$

The number of girls = $(380 - 200) = 180$

In college D:

The total number of students = 440

The number of boys = $(440 + 40)/2 = 240$

The number of girls = $(440 - 240) = 200$

In college E:

The total number of students = 500

The number of boys = $(500 + 60)/2 = 280$

The number of girls = $(500 - 280) = 220$

Colleges	The total number of students	The number of boys	The number of girls
A	480	280	200
B	560	290	270
C	380	200	180
D	440	240	200
E	500	280	220

Answer: A

The total number of students in college F = $480 * 100/80 = 600$

The sum of the number of boys in colleges B and F = $250 * 2 = 500$

The number of boys in college F = $(500 - 290) = 210$

The number of girls in college F = $(600 - 210) = 390$

Hence, the correct answer is option A.

7. Questions

In college A:

The total number of students = 480

The number of boys = $(480 + 80)/2 = 280$

The number of girls = $(480 - 280) = 200$

In college B:

The total number of students = 560

The number of boys = $(560 + 20)/2 = 290$

The number of girls = $(560 - 290) = 270$

In college C:

The total number of students = 380

The number of boys = $(380 + 20)/2 = 200$

The number of girls = $(380 - 200) = 180$

In college D:

The total number of students = 440

The number of boys = $(440 + 40)/2 = 240$

The number of girls = $(440 - 240) = 200$

In college E:

The total number of students = 500

The number of boys = $(500 + 60)/2 = 280$

The number of girls = $(500 - 280) = 220$

Colleges	The total number of students	The number of boys	The number of girls
A	480	280	200
B	560	290	270
C	380	200	180
D	440	240	200
E	500	280	220

Answer: C

The number of boys in college D after transferred = $(240 + 200 * 20/100) = (240 + 40) = 280$

The number of girls in college D after transferred = $(200 + 180/6) = (200 + 30) = 230$

Required sum = $(280 + 230) = 510$

Hence, the correct answer is option C.

8. Questions

In college A:

The total number of students = 480

The number of boys = $(480 + 80)/2 = 280$

The number of girls = $(480 - 280) = 200$

In college B:

The total number of students = 560

The number of boys = $(560 + 20)/2 = 290$

The number of girls = $(560 - 290) = 270$

In college C:

The total number of students = 380

The number of boys = $(380 + 20)/2 = 200$

The number of girls = $(380 - 200) = 180$

In college D:

The total number of students = 440

The number of boys = $(440 + 40)/2 = 240$

The number of girls = $(440 - 240) = 200$

In college E:

The total number of students = 500

The number of boys = $(500 + 60)/2 = 280$

The number of girls = $(500 - 280) = 220$

Colleges	The total number of students	The number of boys	The number of girls
A	480	280	200
B	560	290	270
C	380	200	180
D	440	240	200
E	500	280	220

Answer: B

The number of boys who stay in hostel in college A = $280 * 40/100 = 112$

The number of girls who stay in hostel in college A = $(112 - 50) = 62$

Required sum = $(112 + 62) = 174$

Hence, the correct answer is option B.

9. Questions

In college A:

The total number of students = 480

The number of boys = $(480 + 80)/2 = 280$

The number of girls = $(480 - 280) = 200$

In college B:

The total number of students = 560

The number of boys = $(560 + 20)/2 = 290$

The number of girls = $(560 - 290) = 270$

In college C:

The total number of students = 380

The number of boys = $(380 + 20)/2 = 200$

The number of girls = $(380 - 200) = 180$

In college D:

The total number of students = 440

The number of boys = $(440 + 40)/2 = 240$

The number of girls = $(440 - 240) = 200$

In college E:

The total number of students = 500

The number of boys = $(500 + 60)/2 = 280$

The number of girls = $(500 - 280) = 220$

Colleges	The total number of students	The number of boys	The number of girls
A	480	280	200
B	560	290	270
C	380	200	180
D	440	240	200
E	500	280	220

Answer: A

The sum of the number of girls in colleges A, C, and D together = $(200 + 180 + 200) = 580$

Hence, the correct answer is option A.

10. Questions

In college A:

The total number of students = 480

The number of boys = $(480 + 80)/2 = 280$

The number of girls = $(480 - 280) = 200$

In college B:

The total number of students = 560

The number of boys = $(560 + 20)/2 = 290$

The number of girls = $(560 - 290) = 270$

In college C:

The total number of students = 380

The number of boys = $(380 + 20)/2 = 200$

The number of girls = $(380 - 200) = 180$

In college D:

The total number of students = 440

The number of boys = $(440 + 40)/2 = 240$

The number of girls = $(440 - 240) = 200$

In college E:

The total number of students = 500

The number of boys = $(500 + 60)/2 = 280$

The number of girls = $(500 - 280) = 220$

Colleges	The total number of students	The number of boys	The number of girls
A	480	280	200
B	560	290	270
C	380	200	180
D	440	240	200
E	500	280	220

Answer: D

The total number of students in college E = 500

The difference between the number of girls in colleges C and D = $(200 - 180) = 20$

Required percentage = $(500/20) * 100 = 2500\%$

Hence, the correct answer is option D.

11. Questions

In college A:

The total number of students = 480

The number of boys = $(480 + 80)/2 = 280$

The number of girls = $(480 - 280) = 200$

In college B:

The total number of students = 560

The number of boys = $(560 + 20)/2 = 290$

The number of girls = $(560 - 290) = 270$

In college C:

The total number of students = 380

The number of boys = $(380 + 20)/2 = 200$

The number of girls = $(380 - 200) = 180$

In college D:

The total number of students = 440

The number of boys = $(440 + 40)/2 = 240$

The number of girls = $(440 - 240) = 200$

In college E:

The total number of students = 500

The number of boys = $(500 + 60)/2 = 280$

The number of girls = $(500 - 280) = 220$

Colleges	The total number of students	The number of boys	The number of girls
A	480	280	200
B	560	290	270
C	380	200	180
D	440	240	200
E	500	280	220

Answer: B

The number of boys in college A and B are 280 and 290

$$(2x + 80) = (280 + 290)$$

$$2x + 80 = 570$$

$$2x = 490$$

$$x = 245$$

$$\text{Required value} = (245 * 8) = 1960$$

Hence, the correct answer is option B.

12. Questions

Answer: B

$$\text{The total work} = 45 * 5 * 12 * 100/60$$

$$= 2700 * 100/60$$

$$= 4500 \text{ units}$$

$$\text{The number of days to complete the work} = 4500/(45 * 5) = 20 \text{ days}$$

$$\text{The work completed in 10 days} = 4500 * 20/100 = 900 \text{ units}$$

$$\text{The remaining work} = 4500 - 900 = 3600 \text{ units}$$

$$3600 = 45 * (20 - 10) * (5 + x)$$

$$3600 = 450 * (5 + x)$$

$$8 = (5 + x)$$

$$x = 3$$

Hence, the correct answer is option B.

13. Questions

Answer: E

Let, the age of son = s years

The age of father = ms years

The age of mother = (ms - 20) years

The age of daughter = (ms - 20)/2 years

By using the given data, we cannot find the age of the mother.

So, cannot be determined.

Hence, the correct answer is option E.

14. Questions

Answer: B

The initial quantity of juice in the mixture = $688 * \frac{3}{4} = 516$ litres

The initial quantity of water in the mixture = $688 * \frac{1}{4} = 172$ litres

$$(516 - 64 * \frac{3}{4} + 32) / (172 - 64 * \frac{1}{4} + 4) = 25/x$$

$$(468 + 32) / (156 + 4) = 25/x$$

$$500/160 = 25/x$$

$$20x = 160$$

$$x = 8$$

Hence, the correct answer is option B.

15. Questions

Answer: B

The ratio of the profit share of A to B = $[(x + 1500) * 18 + (0.8x + 1200) * 6] : [(2x + 600) * 18 + (x + 150) * 6]$

$$= (3x + 4500 + 0.8x + 1200) : (6x + 1800 + x + 150)$$

$$= (3.8x + 5700) : (7x + 1950)$$

$$(3.8x + 5700) / (7x + 1950) = 152 / 109$$

$$414.2x + 621300 = 1064x + 296400$$

$$649.8x = 324900$$

$$x = 500$$

Hence, the correct answer is option B.

16. Questions

Answer: B

The speed of stream flow from boat B to boat A means during the journey boat A will travel upstream and B will be downstream.

$$\text{The total length of the river} = [(30 - 10) * 3] + [(20 + 10) * 3]$$

$$= 20 * 3 + 30 * 3$$

$$= 60 + 90$$

$$= 150 \text{ km}$$

Hence, the correct answer is option B.

17. Questions

Answer: A

$$SI = PNR/100$$

$$\text{The amount invested in scheme A} = 40000 * 30/100 = \text{Rs. } 12000$$

$$SI = 12000 * 20 * 2 / 100$$

$$SI = \text{Rs. } 4800$$

$$\text{The amount invested in scheme B} = 40000 * 70/100 = \text{Rs. } 28000$$

$$SI = 28000 * 15/100 * 2$$

$$SI = \text{Rs. } 8400$$

$$\text{Required sum} = (4800 + 8400) = \text{Rs. } 13200$$

Hence, the correct answer is option A.

18. Questions

Answer: C

$$\text{The volume of the cylinder} = \pi r^2 h \text{ cm}^3$$

$$2310 = (22/7) * r^2 * 15$$

$$49 = r^2$$

$$r = 7 \text{ cm}$$

$$\text{The side of the square} = (7 + 3) = 10 \text{ cm}$$

The perimeter of the square = $4a$ cm.

$$= 4 * 10 = 40 \text{ cm}$$

Hence, the correct answer is option C.

19. Questions

Answer: D

Let the number of two numbers be $5x$ and $3x$, respectively.

$$(5x)^2 - (3x)^2 = 1600$$

$$25x^2 - 9x^2 = 1600$$

$$16x^2 = 1600$$

$$x^2 = 100$$

$$x = 10$$

The larger number = $5 * 10 = 50$

Required value = $(50 * 400/100) = 200$

Hence, the correct answer is option D.

20. Questions

Answer: A

Let the cost price of the article = Rs. $100x$

The marked price of the article = $100x * 150/100 = \text{Rs. } 150x$

The selling price of the article = $150x * 85/100 * 80/100 = \text{Rs. } 102x$

The selling price of the article after giving a single discount = $150x * 90/100 = \text{Rs. } 135x$

$$135x - 102x = 99$$

$$33x = 99$$

$$x = 3$$

The cost price of the article = $100 * 3 = \text{Rs. } 300$

Hence, the correct answer is option A.

21. Questions

Answer: C

$$18\% \text{ of } 800 + 42 * 3 - 8^2 = ? * 2$$

$$144 + 126 - 64 = ? * 2$$

$$206/2 = ?$$

$$? = 103$$

Hence, the correct answer is option C.

22. Questions

Answer: E

$$18^2 - 3^3 + 5^2 * 3 + ? = 90 * 4$$

$$324 - 27 + 25 * 3 + ? = 360$$

$$372 + ? = 360$$

$$? = -12$$

Hence, the correct answer is option E.

23. Questions

Answer: B

$$178 + 30 * 8/15 - 20 * 3/4 = ?/2$$

$$178 + 2 * 8 - 5 * 3 = ?/2$$

$$178 + 16 - 15 = ?/2$$

$$178 + 1 = ?/2$$

$$179 * 2 = ?$$

$$? = 358$$

Hence, the correct answer is option B.

24. Questions

Answer: D

$$(256)^{1/2} * 4^{1/2} + 31 * 8 = ?^2 - 9^2$$

$$16 * 2 + 248 = ?^2 - 9^2$$

$$32 + 248 = ?^2 - 81$$

$$280 + 81 = ?^2$$

$$?^2 = 361$$

$$? = 19$$

Hence, the correct answer is option D.

25. Questions

Answer: B

$$12 * 17 + 240 * 5/8 = ? + 190$$

$$204 + 30 * 5 = ? + 190$$

$$204 + 150 - 190 = ?$$

$$? = 164$$

Hence, the correct answer is option B.

26. Questions

Answer: A

$$299.89 * 29.89 - (24.12\% \text{ of } 1199.79) = ? * 2.12$$

$$300 * 30 - 24\% \text{ of } 1200 = ? * 2$$

$$9000 - 288 = ? * 2$$

$$8712/2 = ?$$

$$? = 4356$$

Hence, the correct answer is option A

27. Questions

Answer: C

$$15^2 + 15.12 * 2.87 - ? = 99.12\% \text{ of } 200.21$$

$$225 + 15 * 3 - ? = 99\% \text{ of } 200$$

$$225 + 45 - ? = 198$$

$$270 - ? = 198$$

$$? = 72$$

Hence, the correct answer is option C.

28. Questions

Answer: C

$$1400.01 + \sqrt{1600} - ? = 33.12\% \text{ of } 700.12$$

$$1400 + 40 - ? = 33\% \text{ of } 700$$

$$1440 - ? = 231$$

$$? = 1209$$

Hence, the correct answer is option C.

29. Questions

Answer: D

$$167.12 - 78.09 + 5.12^2 * 4.98 = ?^2 - 11.12$$

$$167 - 78 + 5^2 * 5 = ?^2 - 11.12$$

$$167 - 78 + 25 * 5 = ?^2 - 11$$

$$89 + 125 = ?^2 - 11$$

$$214 + 11 = ?^2$$

$$?^2 = 225$$

$$? = 15$$

Hence, the correct answer is option D.

30. Questions

Answer: E

$$42.12 * 19.12/7 = 8^{1/3} + 156.98 - ?$$

$$42 * 19/7 = 2 + 157 - ?$$

$$114 = 159 - ?$$

$$? = 45$$

Hence, the correct answer is option E.

31. Questions

Answer: E

$$(7 * 2) + 2 = 16$$

$$(16 * 3) + 2 = 50$$

$$(50 * 4) + 2 = 202$$

$$(202 * 5) + 2 = 1012$$

$$(1012 * 6) + 2 = 6074$$

Hence, the correct answer is option E.

32. Questions

Answer: A

$$121 + 31 = 152$$

$$152 + 37 = \mathbf{189}$$

$$189 + 41 = 230$$

$$230 + 43 = 273$$

$$273 + 47 = 320$$

Hence, the correct answer is option A.

33. Questions**Answer: B**

$$(24 - 1) * 1 = 23$$

$$(23 - 1) * 2 = \mathbf{44}$$

$$(44 - 1) * 3 = 129$$

$$(129 - 1) * 4 = 512$$

Hence, the correct answer is option B.

34. Questions**Answer: A**

$$(19 * 0.5) + 0.5 = 10$$

$$(10 * 1) + 1 = 11$$

$$(11 * 1.5) + 1.5 = \mathbf{18}$$

$$(18 * 2) + 2 = 38$$

$$(38 * 2.5) + 2.5 = 97.5$$

Hence, the correct answer is option A.

35. Questions**Answer: B**

$$12^2 + 2 = 146$$

$$13^2 + 2 = 171$$

$$14^2 + 2 = 198$$

$$15^2 + 2 = 227$$

$$16^2 + 2 = 258$$

$$17^2 + 2 = \mathbf{291}$$

Hence, the correct answer is option B.

36. Questions**Answer: E**

$$188 + 12 = \mathbf{200}$$

$$200 + 24 = 224$$

$$224 + 36 = 260$$

$$260 + 48 = 308$$

$$308 + 60 = 368$$

Therefore, 204 is the wrong number

Hence, the correct answer is option E.

37. Questions

Answer: E

$$79 - 9 = 70$$

$$70 * 10 = 700$$

$$700 - 11 = 689$$

$$689 * 12 = 8268$$

$$8268 - 13 = 8255$$

Therefore, 73 is the wrong number

Hence, the correct answer is option E.

38. Questions

Answer: B

$$648/3 = 216$$

$$216/3 = 72$$

$$72/3 = 24$$

$$24/3 = 8$$

Therefore, 76 is the wrong number

Hence, the correct answer is option B.

39. Questions

Answer: E

$$8^3 + 5 = 517$$

$$9^3 + 5 = 734$$

$$10^3 + 5 = 1005$$

$$11^3 + 5 = 1336$$

$$12^3 + 5 = 1733$$

Therefore, 738 is the wrong number

Hence, the correct answer is option E.

40. Questions

Answer: D

1245	1249	1257	1273	1305
4	8	16	32	
	*2	*2	*2	

Therefore, 1259 is the wrong number

Hence, the correct answer is option D.

41. Questions

Answer: B

$$x^2 - 27x + 182 = 0$$

$$x^2 - 13x - 14x + 182 = 0$$

$$x(x - 13) - 14(x - 13) = 0$$

$$(x - 13)(x - 14) = 0$$

$$x = 13, 14$$

$$y^2 - 24y + 143 = 0$$

$$y^2 - 13y - 11y + 143 = 0$$

$$y(y - 13) - 11(y - 13) = 0$$

$$(y - 13)(y - 11) = 0$$

$$y = 13, 11$$

Thus, $x \geq y$

Hence, the correct answer is option B.

42. Questions

Answer: A

$$3x^2 + 32x + 84 = 0$$

$$3x^2 + 18x + 14x + 84 = 0$$

$$3x(x + 6) + 14(x + 6) = 0$$

$$(3x + 14)(x + 6) = 0$$

$$x = -14/3, -6$$

$$y^2 + 15y + 56 = 0$$

$$y^2 + 7y + 8y + 56 = 0$$

$$y(y + 7) + 8(y + 7) = 0$$

$$(y + 7)(y + 8) = 0$$

$$y = -7, -8$$

Thus, $x > y$

Hence, the correct answer is option A.

43. Questions

Answer: D

$$x^2 - 24x + 135 = 0$$

$$x^2 - 15x - 9x + 135 = 0$$

$$x(x - 15) - 9(x - 15) = 0$$

$$(x - 15)(x - 9) = 0$$

$$x = 15, 9$$

$$y^2 - 33y + 272 = 0$$

$$y^2 - 16y - 17y + 272 = 0$$

$$y(y - 16) - 17(y - 16) = 0$$

$$(y - 16)(y - 17) = 0$$

$$y = 16, 17$$

Thus, $x < y$

Hence, the correct answer is option D.

44. Questions

Answer: E

$$x^2 + 17x + 60 = 0$$

$$x^2 + 12x + 5x + 60 = 0$$

$$x(x + 12) + 5(x + 12) = 0$$

$$(x + 12)(x + 5) = 0$$

$$x = -12, -5$$

$$y^2 + 9y + 20 = 0$$

$$y^2 + 4y + 5y + 20 = 0$$

$$y(y + 4) + 5(y + 4) = 0$$

$$(y + 4)(y + 5) = 0$$

$$y = -4, -5$$

Thus, $x \leq y$

Hence, the correct answer is option E.

45. Questions

Answer: A

$$x^2 - 12x + 36 = 0$$

$$x^2 - 6x - 6x + 36 = 0$$

$$x(x - 6) - 6(x - 6) = 0$$

$$(x - 6)(x - 6) = 0$$

$$x = 6, 6$$

$$3y^2 - 19y + 20 = 0$$

$$3y^2 - 15y - 4y + 20 = 0$$

$$3y(y - 5) - 4(y - 5) = 0$$

$$(3y - 4)(y - 5) = 0$$

$$y = 4/3, 5$$

Thus, $x > y$

Hence, the correct answer is option A.